What Changes After SFT? Auditing Hiring Decisions and Internal Pathways

Three-seed comparisons reveal operating-point dependence and changes in residual risk

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1. Problem and protocol

Same qualifications. Different identity cues. Different hiring decisions?

Reduce identity-cue instability while preserving reference agreement.

150 CVs, each paired with 3 jobs = 450 CV–job pairs

3 attribute families; 2 cue styles.
2,700 sets; 31,050 prompts.

Only identity values vary within a set. **Unstable**: both hire and reject.

This poster contributes:

Three-seed operating-point sweeps.
Threshold-controlled overlap of unstable sets.
Cross-stage probes: localization versus editing.

Qwen3.5-9B, English.
LoRA (low-rank adaptation), r32 [1].

SFT: supervised fine-tuning; shared reference.

DPO: direct preference optimization [2].
4,880 on-policy pairs from SFT-v3 sampled mistakes.

GRPO: group relative policy optimization [3].
4,154 tipped prompts; fixed attribute-free decision-reference reward; late LR decay.

Self-distillation: decision-token KL to frozen SFT-v3 on the same pair without identity cues.

Three seeds per added stage.

One CV, 20 gender cues

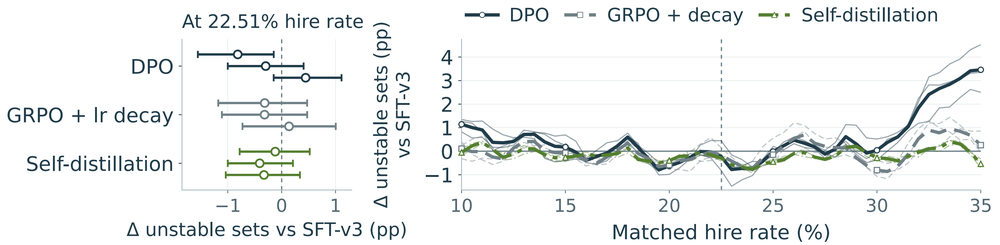
Sales Manager: 3+ years required; CV: 4.5 years.
Base rejects 5 of 20 gender variants.

SFT-v3: all 20 → Hire

2. Behavioural results

SFT delivers the main stability gain

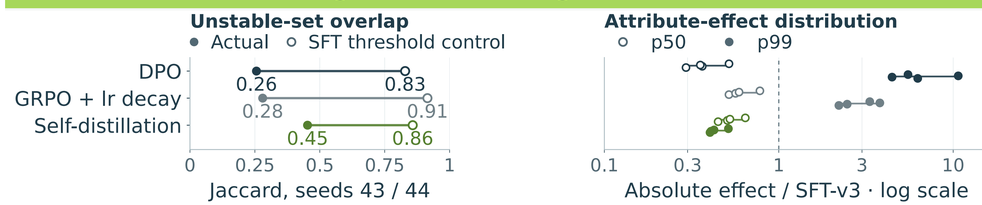
Unstable at 22.51% hire	Base to SFT; paired 95% CI	GPT-4o agreement greedy, own rates
11.8% → 1.8%	**−9.95 [−12.07, −7.83] pp**	**78.8% → 82.4%**



Paired 95% candidate-bootstrap CIs at 22.51% hire.
Seeds 42/43/44 top to bottom; 1 of 9 excludes zero.

Thick: seed means; thin: seeds.
DPO worsens at 31–35% hire.

3. Residual risk changes with training



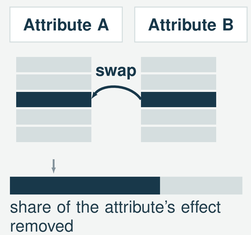
Left: own greedy cuts, seeds 43/44; control recalibrates SFT-v3 to each seed's hire rate.
Right: DPO/GRPO have lower p50 and higher p99 absolute logit effects.
Four contrasts, 450 pairs, one checkpoint per stage; spans are ranges.

95–97% of distillation's unstable sets lie in 35 borderline pairs.
Own thresholds; 35/450 pairs have an absolute attribute-free SFT margin below 2.

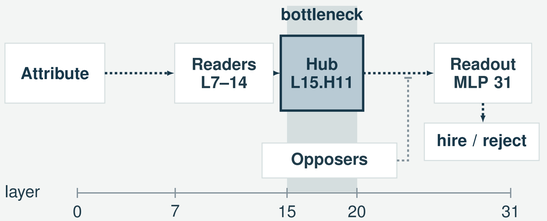
4. Methods and mechanistic findings

A HOW ACTIVATION PATCHING PROBES ATTRIBUTE EFFECTS [5]

Swap one activation



Attribute-sensitive components in the Base model



Readers: layers 7–14 whose patch removes part of the effect upstream. **Hub**: the head whose patch removes the most. **Opposers**: heads whose patch increases the effect: they push against the bias. **Readout**: the last MLP, where the margin forms.

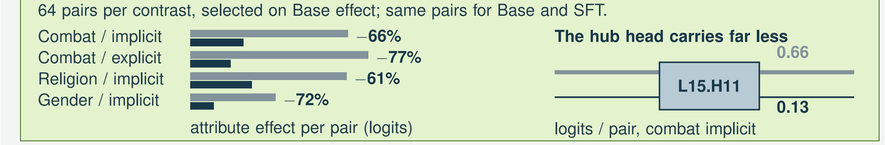
Dashed connections: schematic; edges not individually tested.

Gender / implicit: effect reduction, all 450 pairs

25% 95% CI [12–37%] **vs 72% selected**

B Effect-selected pairs: Base vs SFT-v3

64 pairs per contrast, selected on Base effect; same pairs for Base and SFT.



Post-SFT probes: 64 SFT-selected pairs/contrast, gender 45; one checkpoint per stage.

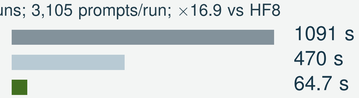
Real component r vs SFT: DPO 0.65–0.89; GRPO 0.90–0.97; distill 0.91–0.98.

Matching trainable parameter counts narrows the observed instability gap [4]
Self-distillation; LoRA restricted to layers 15–20.
11.4M → 58.1M (full: 58.2M): +0.60 → +0.39 pp.
Three-seed means at full-arm hire rates; 1 of 3 paired 95% CIs excludes zero.

5. NVIDIA-accelerated audit

CSCS Alps; NVIDIA GH200; vLLM 0.31; TRL 1.15 / PEFT; PyTorch DDP, bf16

vLLM generation only: ×7.3 vs HF32
Medians of 2 runs; 3,105 prompts/run; ×16.9 vs HF8



Full audit: medians of 4 runs

793 s 0.105 kWh/run

31,050 prompts; gen. 45.65 req/s; NVML GPU energy
Wall includes load/write; scoring excluded.

Training: 1 → 4 GH200

Training: ×3.71 wall; +8% used GPU-h
Training loop; start-up excluded.

Whole step: ×2.67; +50% used GPU-h
Start-up, merge, prep; global batch 32, 40 steps.

Mechanistic audit
Each pass fits one GH200 without offload.

20 min median; 42 GiB max peak
22 runs × 64 pairs; max of sampled GPU peaks.
Margin pass: 32 min median; 87 GiB max peak (31,050 prompts).

SFT delivers the main gain; additional training depends on the operating point and changes residual risk.

One DPO seed alone would have read as a 1 pp gain at its own hire rate. Exploratory development benchmark; agreement is a proxy.

[1] Hu et al., LoRA: Low-Rank Adaptation of Large Language Models (2021). [2] Rafailov et al., Direct Preference Optimization: Your Language Model is Secretly a Reward Model (2023). [3] Shao et al., DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models (2024).
[4] Hase et al., Does Localization Inform Editing? NeurIPS (2023). [5] Zhang & Nanda, Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. ICLR (2024).

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3

Results + code